

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 7B nr. 1.4

$$w = \ln(u + v + z)$$

$$u = \cos^2 t, v = \sin^2 t, z = t^2$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial u} \frac{du}{dt} + \frac{\partial w}{\partial v} \frac{dv}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

$$\frac{\partial w}{\partial u} = \frac{1}{u + v + z}$$

$$\frac{\partial w}{\partial v} = \frac{1}{u + v + z}$$

$$\frac{\partial w}{\partial z} = \frac{1}{u + v + z}$$

$$\frac{du}{dt} = -2 \cos t \sin t$$

$$\frac{dv}{dt} = 2 \sin t \cos t$$

$$\frac{dz}{dt} = 2t$$

$$\frac{dw}{dt} = \frac{1}{u + v + z} (-2 \cos t \sin t) + \frac{1}{u + v + z} (2 \sin t \cos t) + \frac{1}{u + v + z} (2t)$$

$$\frac{dw}{dt} = \frac{2t}{u + v + z}$$

$$\frac{dw}{dt} = \frac{2t}{\cos^2 t + \sin^2 t + t^2}$$

$$\frac{dw}{dt} = \frac{2t}{1 + t^2}$$

References